

CHANG YAO HSIEN (VICTOR)

SENIOR FULL-STACK ENGINEER | REACT · TYPESCRIPT · GO · REAL-TIME SYSTEMS

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PROFILE

Senior Full-Stack Software Engineer with 7+ years of experience building production web platforms across frontend, backend, and cloud infrastructure. **Strong in React/TypeScript and Go/NestJS**, with experience designing **real-time workflows, REST/gRPC APIs, event-driven systems, PostgreSQL/Redis, AWS, Docker, and Kubernetes**. Proven end-to-end ownership from requirements and architecture through implementation, deployment, and production troubleshooting, collaborating with Product Managers and distributed engineering teams.

EXPERIENCE

Cypherlab Sdn. Bhd.

Feb 2026 - Aug 2026

Senior Full-Stack Engineer

Real-time sports and e-sports betting backoffice platform

Tech Stack: TypeScript, React, Go, gRPC, NATS JetStream, WebSocket, PostgreSQL, Redis, ClickHouse, Docker, Kubernetes

- Built **React and TypeScript backoffice workflows** for live sports and e-sports operations while contributing to a **domain-oriented modular BFF in Go**, integrating sports-manager HTTP APIs with order-book, wallet, and user-service via gRPC.
- Modularized **WebSocket message handling** and built a **NATS JetStream -> in-memory projections -> WebSocket** pipeline, handling event ordering and stale updates to maintain consistent real-time market and odds state.
- Delivered operator controls for **market closure and odds overrides** and implemented a shared **Command Gate** for high-risk actions with operation reasons, idempotency, maker-checker validation, and audit trails.
- Separated API serving, ClickHouse ingestion, and settlement processing into independent Go binaries deployed as Kubernetes workloads, isolating synchronous API traffic from asynchronous processing.

Mediconcen

Feb 2022 - Jan 2026

Senior Full-Stack Engineer

Hong Kong-based InsurTech delivering digital health insurance solutions

Tech Stack: TypeScript, React, NestJS, Go, PostgreSQL, Redis, AWS, Docker, Kubernetes

- Modernized a legacy **React** codebase from class components to functional components with Hooks and implemented **infinite scrolling and list virtualization** for large data-heavy views.
- Owned end-to-end clinic and insurer workflows covering **eligibility verification, copayment calculation, claims processing, and third-party insurer APIs**, from requirement clarification through production support.
- Modernized legacy backend services from **PHP to NestJS**, introducing ORM-based data access and DTO validation, and added **Redis caching** to reduce repeated database reads and query load.
- Migrated production workloads from **AWS ECS to Kubernetes** and troubleshooted issues across frontend, backend services, databases, and third-party APIs with Product Managers and cross-country engineering teams.

Paradromix

Jul 2021 - Feb 2022

Frontend Engineer

Government-grade crypto management systems and enterprise web platforms

Tech Stack: TypeScript, Vue, Nuxt (SSR), AWS, Nginx, Docker

- Led frontend delivery for SSR web platforms and built a **custom CMS** for non-technical teams, with focus on SEO, maintainable UI architecture, and production deployment.

ULIC TEK

Jan 2018 - Jul 2021

Software Engineer

Image processing systems and industrial desktop-web hybrid solutions

Tech Stack: Vue, TypeScript, Angular, C++, MFC, OpenCV

- Built image-processing functionality in **C++/OpenCV** and modernized legacy desktop applications with web-based UI layers for industrial desktop-web hybrid solutions.

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, Next.js, Vue, Nuxt, SSR, SEO, WebSocket, Component Architecture

Backend: Go, Node.js, NestJS, Express, gRPC, REST APIs, WebSocket, NATS JetStream

Data: PostgreSQL, MySQL, Redis, ClickHouse, MongoDB

Cloud & Architecture: AWS, Docker, Kubernetes, Microservices, Event-Driven Architecture, CI/CD, Linux, Nginx

Languages: English, Mandarin, Basic Japanese

EDUCATION

Bachelor of Chemistry, Chia Nan University of Pharmacy and Science

Graduated 2017